

TABLE OF CONTENTS

E-001 GENERAL ELECTRICAL SPECIFICATION

	Page No.
1. SCOPE OF WORK	E-001-2
2. CODES AND STANDARDS	E-001-2
2.1 National	E-001-2
2.2 International Electrotechnical Commission.....	E-001-2
2.3 National Electrical Manufacturers Association	E-001-4
2.4 National Fire Protection Association.....	E-001-4
3. OPERATING AND DESIGN CONDITIONS.....	E-001-4
3.1 Power Supply.....	E-001-4
3.2 Control Voltages.....	E-001-5
3.3 Environmental Considerations	E-001-5
4. DETAILED REQUIREMENTS	E-001-6
4.1 Ratings.....	E-001-6
4.2 Current Transformers	E-001-6
4.3 Voltage Transformers	E-001-7
4.4 Busbars and Connections	E-001-8
4.5 Grounding (Earthing)	E-001-8
4.6 Insulators	E-001-8
4.7 Electrical Insulation.....	E-001-9
4.8 Insulating Oil.....	E-001-9
4.9 Oil Filled Chambers	E-001-9
4.10 Anti-Condensation Space Heaters	E-001-10
4.11 Cable Boxes and Glands.....	E-001-10
4.12 Junction Boxes.....	E-001-10
4.13 Protection Equipment and Relays.....	E-001-11
4.14 Power Panels and Devices	E-001-12
4.15 Control Panels and Devices	E-001-15
4.16 Ratings of Electrical Contacts	E-001-16
4.17 Solenoid Valves.....	E-001-17
4.18 Painting.....	E-001-17
4.19 Electrical Consumer Lists (ECL)	E-001-17
5. PERFORMANCE TESTING AND GUARANTEES.....	E-001-17
6. QUALITY ASSURANCE/QUALITY CONTROL.....	E-001-17

E-001 GENERAL ELECTRICAL SPECIFICATION

1. SCOPE OF WORK

This Specification outlines the general electrical requirements for the equipment included in this Contract. Refer to Part 4, Subsection 1.2, ‘Summary of Work’ and the individual Technical Specifications for details.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards” and the following specific standards.

2.1 National

PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation Indonesia).

SPLN : Standard Perusahaan Umum Listrik Negara

2.2 International Electrotechnical Commission

IEC 60028 : International standard of resistance for copper

IEC 60044 : Instrument transformers

IEC 60050 : International Electrotechnical Vocabulary

IEC 60051 : Direct acting indicating analogue electrical measuring instruments and their accessories

IEC 60059 : IEC standard current ratings

IEC 60071 : Insulation co-ordination

IEC 60073 : Basic and safety principles for man-machine interface, making and identification-Coding principles for indicators and actuators

IEC 60076 : Power transformers

IEC 60079 : Electrical apparatus for explosive gas atmospheres

IEC 60141 : Test on oil filled and gas pressure cables and their accessories

IEC 60146 : Semiconductor converters

IEC 60183 : Guide to the selection of high voltage cable

IEC 60186	:	Voltage transformers
IEC 60228	:	Conductors of insulated cables
IEC 60255	:	Electrical relays
IEC 60265	:	High-voltage switches
IEC 60269	:	Low-voltage fuses
IEC 60282	:	High-voltage fuses
IEC 60296	:	Fluids for electrotechnical applications – Unused mineral insulating oils for transformers and switchgear
IEC 60309	:	Plugs, socket outlets and couplers for industrial purposes
IEC 60331	:	Tests for electric cables under fire conditions - Circuit integrity
IEC 60423	:	Conduits for electrical purpose - Outside diameters of conduits for electrical installations and threads for conduits and fittings.
IEC 60439	:	Assemblies low voltage switchgear and control gear.
IEC 60445	:	Basic and safety principles for man-machine interface, marking and identification - Identification of equipment terminals and of terminations of certain designed conductors, including general rules for an alphanumeric system.
IEC 60446	:	Basic and safety principles for man-machine interface, marking and identification - Identification of conductors by colors or numerals
IEC 60447	:	Man-machine-interface (MMI) – Actuating principles
IEC 60502	:	Power cables with extruded insulation and their accessories for rated voltages from 1 kV ($U_m = 1.2$ kV) up to 30 kV ($U_m = 36$ kV)
IEC 60529	:	Degrees of protection provided by enclosures (IP code)
IEC 60605	:	Equipment reliability testing.

IEC 60614	:	Conduits for electrical installations Specification.
IEC 60617	:	Graphical symbols for diagrams
IEC 60811	:	Common test methods for insulating and sheathing materials of electric cables and optical cables.
IEC 60947	:	Low-voltage switchgear and control gear
IEC 61082	:	Preparation of documents used in eletrotechnology.
IEC 62271	:	High-voltage switchgear and control gear

2.3 National Electrical Manufacturers Association

NEMA ICS 2 : Industrial Control and Systems: Controllers, Contactors, and Overload Relays Rated 600 Volts.

2.4 National Fire Protection Association

NFPA 70 : National Electrical Codes

3. OPERATING AND DESIGN CONDITIONS

3.1 Power Supply

Auxiliary electric power shall be distributed from distribution equipment (medium-voltage switchgear, low-voltage switchgear, motor control centers, etc.) in the power plant for normal operation of the plant auxiliaries. The power supply voltages shall be as detailed below. The Contractor shall select and size his electrical equipment accordingly for PLN approval.

6 kV $\pm 10\%$, 3 phase, for motors > 150 kW and for PDC transformer loads, fed from 6.3 kV switchgear.

400 V $\pm 10\%$, 3 phase, for motors > 60 kW and < 150 kW, fed from 400 V switchgear (PDCs).

400 V $\pm 10\%$, 3 phase, for motors (including motor operated valves) and for loads up to 60 kW fed from 400 V MCCs.

230 V AC $\pm 10\%$, for single phase motors up to 750 W (from 230 V AC distribution panels).

230 V AC supply shall be used for space heaters, enclosure interior lighting, ventilation fans, convenience outlets, etc.

The supply frequency for continuous operation shall be 50 Hz + 1% to - 2%. Transient variations may be $\pm 5\%$.

220 V DC nominal, for DC motors, DC motor operated valves (MOVs) and control. The DC supply voltage range shall be 187 to 242 V (- 15%, + 10%).

24 V DC nominal, for control equipment loads from the central control building. The DC control equipment supply voltage range shall be 21.6 to 26.4 V, if required.

230 V $\pm 2\%$, 50 Hz $\pm 0.5\%$, regulated single phase uninterruptible power supply shall be available for vital/critical plant instrumentation and control electrical loads.

The Contractor shall furnish any voltage regulating equipment required for proper operation of his equipment. The Contractor shall supply an alarm relay and contact to alarm loss of or under-voltage condition in any power supply conversion unit or voltage regulating equipment.

3.2 Control Voltages

Control voltages shall be 220 volts DC for medium-voltage and low-voltage switchgear unless otherwise approved

DC auxiliaries and control power for substation shall be 220V nominal voltage and with range of 188-242 volt for closing and 154-242 volt for tripping.

DC auxiliaries and control power for instrumentation shall be 24 V nominal voltage with range of 21.6-26.4 volt, if required.

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3.3 Environmental Considerations

The electrical equipment shall be suitable for operation in a combined cycle plant on a tropical seacoast.

The electrical equipment in hazardous areas as defined by IEC 60079 shall comply with the requirements of IEC 60079 or equivalent.

4. DETAILED REQUIREMENTS

4.1 Ratings

4.1.1 Normal Current Ratings

Every current carrying part of the equipment including breakers, contactors, switches, isolators, tapping switches, fuses, busbars, current transformers, connections and joints shall be capable of carrying their rated current continuously under the design parameters, as specified, and in no case shall the permitted temperature rise be exceeded.

4.1.2 Temperature Rise

The limits of temperature rise of all equipment shall be determined in accordance with the approved standards, and where applicable, shall be reduced such that the final attained temperatures, based on the equipment ambient at the Site, do not exceed maximum allowable equipment temperatures.

4.2 Current Transformers

Current transformers used for instruments shall be of accuracy Class 0.5, or as specified in individual Technical Specifications. Current transformers shall comply with the requirements of IEC 60044.

Current transformers for combined purposes (e.g. inverse time relays and indicating meters) shall have dual accuracy Class 5P20/Class 0.5 performances.

Current transformers for tariff metering and for guaranteed performance tests shall be of accuracy Class 0.2.

The rated volt-ampere output of each current transformer shall not be less than the connected burden in service. The burden of cable connections shall be taken into account. The minimum volt-ampere output rating shall be 30 VA. Volt ampere less than this figure shall be supported with the approved calculation.

Current transformers shall be capable of withstanding, without damage, the peak and rated short-time currents of their associated equipment. Standard secondary current will be 5 or 1 amperes unless otherwise approved by the Owner.

The secondary windings of each set of current transformers shall be grounded at one (1) point only via a bolted disconnecting link. The transformer core, where accessible, shall be separately grounded.

Where multi-ratio secondary windings are specified, a label shall be provided at the secondary terminals of the current transformer indicating clearly the connections required for each tap. The connections and the ratio in use shall be indicated on all connection diagrams.

Design magnetization curves and DC resistance values shall be submitted for each current transformer to be used for protective relaying purposes and shall be subsequently verified by routine factory tests.

4.3 Voltage Transformers

Voltage transformers shall comply with the requirements of IEC 60044 and 60186 and shall be of Class 1.0 accuracy or as specified. The VA output shall be 50% in excess of the design requirements unless otherwise specified. The minimum volt-ampere output rating shall be 100 VA. Volt ampere less than this figure shall be supported with approved calculation.

Voltage transformers for performance test and measurement for auxiliary power consumption and metering shall be of Class 0.2 accuracy.

Secondary fuses or MCB's shall be provided, as close as possible, to each voltage transformer and labeled to show their function and phase color. The secondary circuit shall be monitored immediately after these fuses to detect and alarm individual fuse failure.

Voltage transformers shall be designed so that saturation of their cores does not occur when 1.732 times normal voltage is applied to each winding.

Magnetization curves for each type of voltage transformer shall be submitted for approval.

The standard secondary voltage between phases shall be 110 volts unless special circumstances dictate otherwise and are approved by the Owner.

Secondary circuits shall not be connected in parallel in normal operation.

4.4 Busbars and Connections

Busbars and connections shall comply with the applicable standard and shall be of electrolytic copper or aluminum. Medium-voltage busbar assemblies and connections shall be insulated.

Medium-voltage busbars shall be contained in a separate compartment within the general casing of switchgear and fused disconnect switches. Busbar barriers shall be provided, if required, between switchgear equipment to prevent the spreading of ionized gases in the event of a fault. Phase colors shall be Red (Phase “R”) - Yellow (Phase “S”) - Black (Phase “T”) and shall be arranged in accordance with IEC 60445.

Access to busbars and the connections directly connected thereto shall be gained only by the removal of covers secured by bolts or screws. Such covers shall be clearly and indelibly marked ‘BUSBARS’.

All busbars shall be of copper material and joints are to be silver-plated.

4.5 Grounding (Earthing)

All metal parts, other than those forming part of an electrical circuit, shall have provisions for connection, in an approved manner, to separate copper ground bars in the control panels, distribution panels, electrical enclosures, etc.

These ground bars shall not less than 7 x 50 mm² cross section for the distribution panels or electrical enclosures containing power conductors and terminals and 7 x 25 mm² cross section for the control panels.

4.6 Insulators

Porcelain insulators shall be of best quality porcelain. The clamping surfaces of all porcelain insulators shall be ground accurately and shall be free of glaze.

Each porcelain insulator shall bear the manufacturer's mark and batch identification which shall be applied before firing.

Insulators shall be mounted and the method of attaching connection shall be such that there is no likelihood of their being mechanically overstressed during normal tightening of the mounting and connection fixings. Similar provision shall be made to accommodate expansion and contraction of the connections due to the temperature likely to be attained during fault conditions.

The arrangement of all insulators shall be such as to minimize the accumulation of dust.

4.7 Electrical Insulation

Linseed oil and untreated materials of fiber, leatheroid or other similar hygroscopic materials shall not be used for insulation purposes. Untreated leatheroid or other materials may be used for mechanical protection of winding insulation. Varnish with a linseed oil base shall not be used.

No material of a hygroscopic nature shall be used for covering coils. Where inter-leaving between windings in coils is necessary, only the best manila paper, thoroughly dried, which permits penetration by the insulating varnish or wax, shall be used.

4.8 Insulating Oil

Insulating oil shall be "Shell Diala Oil B" comply with IEC 60296, or equivalent.

Oil shall be provided by the Contractor for all oil filled apparatus.

Devices and/or equipment which are located within a building shall utilize a dry type, non flammable insulation or approved equal. If an oil insulated equipment is located within 3.5 meters of a building wall, the the building wall shall be designed as a fire wall.

4.9 Oil Filled Chambers

All joints of oil-filled fabricated chambers, other than those which have been broken, shall be welded and care shall be taken to ensure that the chambers

are oil-tight.

Oil filling level indication shall be engraved on the inside and outside of chambers.

4.10 Anti-Condensation Space Heaters

All switchboards, panels, cubicles and the like, except in air conditioning room, shall incorporate electric heaters capable of providing movement of sufficient heated air to prevent condensation. The heaters shall be wired internally to the equipment terminal blocks for connection to the power supply. The heaters voltages shall be operated at 220 V AC.

The operation of the heaters shall be thermostatically controlled; however, the heaters shall be suitable for continuous operation.

4.11 Cable Boxes and Glands

Electrical equipment supplied under this Contract shall be fitted with all necessary cable termination boxes which shall be complete with all required fittings.

Gland plates and boxes shall be provided with sufficient internal space left to accommodate mechanical compression type glands and locking nuts.

Flexible connections shall be provided between cable lugs and bushings for all cables 300 mm² and larger.

For currents in excess of 500 A, adequate steps shall be taken to minimize the effects of eddy currents in the gland and bushing-mounting plate. All glands shall be provided with a ground tag or equal facility

Where insulated glands are specified, removable connections for bonding across the gland insulation shall be provided.

4.12 Junction Boxes

The Contractor shall provide local junction boxes on or near his equipment as specified in the equipment specifications or as required for convenient field or external wiring of the equipment.

All junction boxes shall be of substantial galvanized sheet steel construction, fitted with external fixing lugs and finished in accordance with the specification requirements for painting and finishing. Outdoor junction boxes shall be water tight, gasketed and hot dipped galvanized with a degree of protection of IPW 54 of IEC 60529. Other indoor junction boxes shall be dust tight, gasketed with a degree of protection IP 42 of IEC 60529.

Junction boxes located in hazardous explosive gas areas shall be explosion proof and shall comply with IEC 60079 or equivalent requirements.

Separate junction boxes shall be provided for all low level analog signals.

4.13 Protection Equipment and Relays

Protective relays shall comply with the requirements of IEC 60255, or better. They shall be preferably of the digital multifunction type relay. In case of individual electromagnetic or static type relay, these shall be of the withdrawable type with provision for inserting a test plug from the front between sets of contact fingers. The test plug shall be provided with a flexible cable for connection to a portable relay test set. Any relay not of the withdrawable type or where it is not possible to insert a test plug between contact fingers, it shall be provided with a test socket, adjacent to the relay, to enable the relay to be tested in service. This requirement is additional to any permanently connected injection test scheme.

The relay cases shall be arranged to short current transformer leads and open the trip circuit when the relay is withdrawn. Metal cases shall be grounded.

Two (2) test plugs to suit each different type of relay case or test socket shall be provided for each relay panel. The test plug shall have terminals for both the relay and wiring side connections which will accept both wires and plug connectors, and be complete with lengths of flexible cable for connection to a portable relay test set.

All relays which are connected to complete, either the tripping circuit of a circuit breaker, or the coil circuit of an auxiliary tripping relay, shall be provided with operation indicators which, whenever possible, shall be of the mechanically operated type. Indicators shall also be provided on such additional relay elements as will enable the phase of the fault condition to be

identified as well as the exact tripping function which operated. Each indicator, whether of the electrically operated or mechanically operated kind, shall be capable of being reset by hand without opening the relay case. It shall not be possible to operate the relay when resetting the operating indicator. Each indicator shall be so designed that it cannot show before the relay has completed its operation. Indicators shall not reset during a failure of auxiliary power to the relay. Where relays are required to operate with a fixed time delay, the time delay shall not be of the dashpot type.

Direct current relays, associated with protection, shall be rated for 220 V DC.

Each auxiliary relay for external interface shall be provided with a minimum of two (2) normally-open and two (2) normally- closed electrically independent contacts or two (2) interchangeable spare contacts for the Owner's use.

Settings for protective relays which require discriminative time grading and discriminative protection scope with other protection schemes shall also be determined by the Contractor.

The Contractor shall be responsible for providing to the Owner all relay setting details for relays within his supply. The Contractor shall calibrate and set all relays during installation and commissioning.

All calculations to determine relay setting and CT and VT requirements including overcurrent and ground fault current coordination shall be submitted to the Owner for approval.

4.14 Power Panels and Devices

4.14.1 General

The Contractor shall furnish specified power cabinets, cubicles, starter panels, and local starters, in addition to those which are integral part of his equipment.

Minimum size of the motor starters and contactors shall be NEMA size 1 or IEC equivalent. NEMA size 1 AC combination starters shall have field adjustable instantaneous trips (magnetic only) in their molded case circuit

breakers. For NEMA size 2 and larger starters, the breakers shall be thermal magnetic.

The power panels shall have a degree of protection, as defined by IEC 60529, of IPW 55 for outdoor or IP 42 for indoor locations.

All power connections shall be shielded to prevent accidental contact by personnel.

The circuit breakers shall have a manual operating handle on the front of their enclosure door. The position of the handle shall indicate whether the breaker is open or closed.

The circuit breaker and operating devices on the doors shall be arranged to prevent opening or closing the doors unless the circuit breaker and operating devices are in the “off” position. A defeater shall be provided to permit opening of the door when the breaker is in the “on” position.

An individual 380/110 volt AC, 50 Hz, two (2) winding control transformer connected on the load side of the air circuit breaker shall be provided for each starter unless the manufacturer’s standard control requires a common control transformer or specified otherwise. The secondary of the control transformer shall have one (1) side grounded and the other side fused on both sides.

4.14.2 Power Panel Devices

4.14.2.1 Combination Starters

Each combination starter shall consist of a circuit breaker, and a magnetic starter, and mounted in the same compartment. Reversing and two-speed starters shall consist of a circuit of a circuit breaker and two (2) magnetic starters with mechanical and electrical interlocks to prevent simultaneous closing.

4.14.2.2 Circuit Breakers

Air circuit breakers for individual feeder circuits shall be 600 volt AC, 50 Hz, 2 or 3-pole for AC service; 250 volt, 2-pole for DC service, manually operated and trip free with inverse-time thermal-overload and instantaneous overcurrent trips on all poles. The instantaneous trip shall operate at approximately ten (10) times rated current. The thermal element shall be protected against fusing or damage on short circuit.

Air circuit breakers for motor starters shall conform to the above except that the inverse-time thermal overload trip shall be omitted and the instantaneous trip shall be adjustable to fifteen (15) times the motor current.

4.14.2.3 Magnetic Starters

Magnetic starters shall be rated 600 volt AC, 50 Hz, 3-pole or 250 volt DC, 2-pole single throw contactors with external hand reset overload relays for each pole and current ratings as appropriate for the motor sizes.

Each contactor shall have a normally-open (open when contactor coil is de-energized) auxiliary contact for sealing-in and under-voltage protection and in the case of reversing or two-speed contactors, shall have a normally-closed (closed when contactor coil is de-energized) auxiliary contact for electrical interlock. Auxiliary contacts specified shall be in addition to the above. A 110 volt (not 380 volt) auxiliary relay shall be used when the number of auxiliary contacts specified exceeds the number permitted by the manufacturer's design.

Starters shall be wired for three-phase AC operation. Control leads and leads from extra auxiliary contacts shall be connected to terminal blocks.

4.14.2.4 Fuses and Links

Fuse and link holders and bases shall be in accordance with IEC 60269 and 60282, as applicable. Fuse links, fuse contacts and fixed contacts shall be shielded to prevent inadvertent contact with live metal when the fuse is being inserted or withdrawn.

Fuses shall be of the non-deteriorating HRC (high rupturing capacity) type and fuse bridges shall be so designed as to permit detection of fuse-operation

without removing the bridge. Test links in current transformer circuits shall be of plug-in type.

The fuses shall be identified and labeled. Approved codes symbols shall be used on diagram and on fuse or link labels.

Fused switches for three-phase motors shall have tripping devices to prevent single phasing of motors when a fuse actuates.

4.15 Control Panels and Devices

4.15.1 Control Panels

All control panels shall be in accordance with Technical Specification I-002 Local Control Panel.

4.15.2 Control Panel Devices

4.15.2.1 Indicating Lamps

Indicating lamps including LED shall be complete with bulbs, translucent lenses (engraved and/or colored as applicable) and colored screens (if applicable). Bulbs and LED shall be of long life type and shall be operated at approximately 90% of their rated voltage. Suitable voltage dropping resistors shall be provided, if required, to reduce the voltage applied to the bulbs and LEDs.

Every circuit breaker and contactor panel shall be equipped with one (1) red and one (1) green indicating lamp indicating, respectively, circuit “closed” and circuit “open” and an amber lamp, where required, for indicating “auto-trip”.

Every circuit breaker panel shall be provided with a trip circuit supervision relay, where specified, to indicate and alarm when the trip circuit is faulty.

Indicator lamps for motor or solenoid actuated valves shall be red for open and green for fully closed. Fuses shall be provided for indication circuits.

4.15.2.2 Auxiliary Switches

Auxiliary switch contacts shall have a positive wiping action when closing. All auxiliary contacts shall be wired to terminal blocks.

Auxiliary switches shall be provided to interrupt the supply of current to the trip coil of each circuit breaker or contactor immediately after the breaker or contactor has opened. Auxiliary switch contacts shall be rated 0.5 A (inductive) at 250 V rated DC.

4.15.2.3 Control Switches

Control switches shall be in accordance with IEC 60947 or equivalent and shall have a mechanical endurance rating of 1 million operations.

Switches shall be arranged to operate clockwise when closing circuit breakers and anti-clockwise when opening them.

The switch shall be equipped with a mechanical target to indicate the last position of the circuit breaker. The close target shall be red and the open target green.

Local/remote control selector switches shall be arranged for locking in both positions.

4.15.2.4 Pushbuttons and Pushbutton Stations

Pushbuttons shall be heavy duty, oil tight, dust tight and weatherproof type suitable for panel, wall or bracket mounting.

Pushbuttons shall be shrouded or recessed to prevent accidental operation. Start pushbuttons shall be red and stop buttons colored green. Emergency stop pushbuttons shall be colored red and shall be protected or guarded to prevent inadvertent operation.

4.16 Ratings of Electrical Contacts

All electrical contacts for external interfaces, pressure switches, limit switches, temperature switches, etc. shall be rated not less than 0.5 A (inductive) at 250 V DC and 5 A at 220 V AC, except where specified otherwise in this Contract.

4.17 Solenoid Valves

Solenoid valves shall have encapsulated coils with Class H insulation and be rated for continuous energization duty.

4.18 Painting

All equipment shall be painted in accordance with the specifications and requirements of applicable Civil Technical Specifications.

4.19 Electrical Consumer Lists (ECL)

ECL shall be provided by the Contractor in the form and format to be advised by the Owner.

5. PERFORMANCE TESTING AND GUARANTEES

The Contractor shall perform all tests necessary to demonstrate the suitability of his equipment. All test reports and certificates shall be subject to the approval of the Owner. Type test report shall be submitted incase the type test of identical equipment already performed and therefore no repeated type test is required.

After erection, the Contractor shall demonstrate the proper operation of the equipment. Special attention shall be given to test the equipment in all operating modes.

6. QUALITY ASSURANCE/QUALITY CONTROL

The Contractor shall fulfill the quality assurance/quality control requirements listed for individual equipment in the respective equipment specifications and in Part 4, Section 2, Subsection 2.7.

The quality assurance/quality control procedures shall include, but not be limited to, the followings:

- (1) Design criteria and engineering calculations
- (2) Factory and field inspections
- (3) Wiring insulation testing
- (4) Control and functional tests
- (5) Protective relay settings and tests

The quality assurance/control documentation shall include, but not be limited to, the following:

- (1) Inspection Reports
- (2) Certified Test Reports
- (3) Material certification for the specified requirements of cables, insulation, conductor material, etc.

(Last page of E-001)